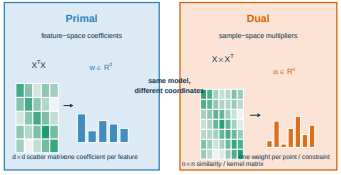
Optimization in Machine Learning

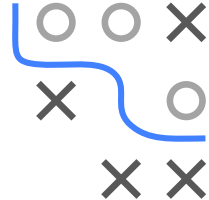
Constrained Optimization
Primal Versus Dual Methods



Learning goals

- Compare primal and dual viewpoints for the same constrained problem
- Understand why dual methods are not automatically superior or required for kernels
- Relate computational cost to the number of variables d and data points n
- See why approximate primal solutions are often easier to interpret and use

TWO VIEWPOINTS FOR THE SAME PROBLEM



- The primal solves the original constrained problem

$$\min_{\boldsymbol{w} \in \mathcal{S}} F(\boldsymbol{w}),$$

  directly in the model parameters $\boldsymbol{w}$.

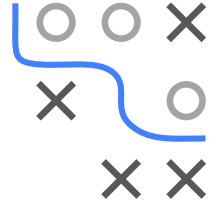
- The dual solves

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} L(\boldsymbol{\alpha}), \qquad L(\boldsymbol{\alpha}) = \min_{\boldsymbol{w}} H(\boldsymbol{w}, \boldsymbol{\alpha}),$$

  in the multiplier variables $\boldsymbol{\alpha}$.

- Thus, primal and dual methods come from the same Lagrangian, but they optimize in different coordinates.
- Primal iterates live in the original parameter space; dual iterates live in multiplier space and usually need a recovery step back to $\boldsymbol{w}$.
- The main practical differences are dimension, constraint structure, computational cost, and interpretability.
- This final section is a decision guide: when does staying primal buy more than reformulating?

DUALITY IS A REFORMULATION



- The dual is created from the same Lagrangian as before:

$$H(\boldsymbol{w}, \boldsymbol{\alpha}) = F(\boldsymbol{w}) + \sum_{i=1}^{m} \alpha_i g_i(\boldsymbol{w}), \qquad \boldsymbol{\alpha} \geq \mathbf{0}.$$

- Passing from primal to dual does **not** remove optimization difficulty; it changes the coordinates of the problem.
- In practice, deriving the dual often introduces one multiplier per constraint, extra sign or box constraints on these multipliers, and a mapping back from $\boldsymbol{\alpha}$ to $\boldsymbol{w}$.
- Hence the relevant question is not “primal or dual in general?”
- The relevant question is:

Which representation best matches the dimension, constraints, and approximation needs of the current problem?

TOY EXAMPLE

Reuse

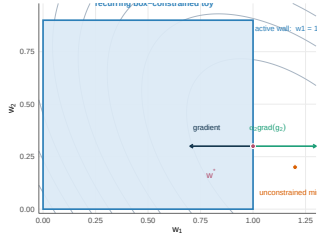
$$\min_{\boldsymbol{w}} F(\boldsymbol{w}) = (w_1 - 1.2)^2 + (w_1 - 1.2)(w_2 - 0.2) + (w_2 - 0.2)^2$$

$$\text{s.t.} \qquad 0 \le w_1 \le 1, \qquad 0 \le w_2 \le 0.9.$$

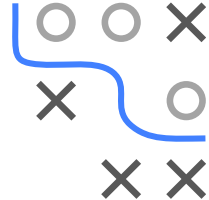
- Primal view: 2 variables, cheap clipping, direct feasible iterates.
- Dual view: 4 nonnegative multipliers, one for each box wall.
- At the solution, only one multiplier is nonzero:

$$\boldsymbol{\alpha}^* = (0, 0.3, 0, 0)^\top.$$

- This is exactly the kind of low-dimensional problem where the primal view stays simpler than the reformulation.



KERNEL METHODS: DUAL HELPFUL, NOT ESSENTIAL



Linear SVM in the primal

$$\min_{\boldsymbol{w},b} \quad \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^{n} \max\left\{1 - y^{(i)}\left(\boldsymbol{x}^{(i)\top}\boldsymbol{w} + b\right), 0\right\}$$

Soft-margin SVM in the dual

$$\max_{\boldsymbol{\alpha}} \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y^{(i)} y^{(j)} k_{ij}$$

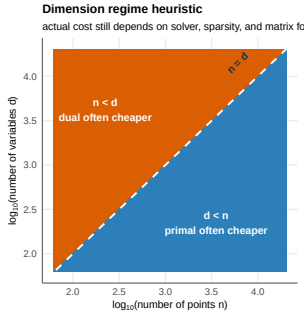
$$\text{s.t.} \quad 0 \le \alpha_i \le C, \qquad \sum_{i=1}^{n} \alpha_i y^{(i)} = 0$$

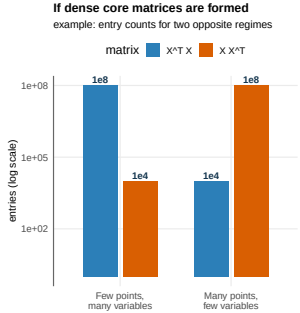
- The dual depends only on similarities k_{ij}, so replacing dot products by a kernel is immediate.
- But the primal SVM is already a valid unconstrained optimization problem once the hinge loss is written inside the objective.
- Therefore, kernel methods require similarities, not duality itself; for many kernel regularization problems, the representer theorem shows that the solution lies in the span of training-point similarities.
- Duality is useful here, but it is **not** the only reasonable way to solve or kernelize the model.

WHICH SIDE IS COMPUTATIONALLY CHEAPER?

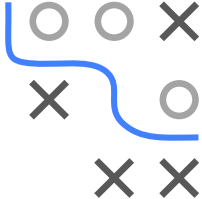
Primal side: updates in parameter space $\boldsymbol{w} \in \mathbb{R}^d$
Dual / kernel side: updates in multiplier space $\boldsymbol{\alpha} \in \mathbb{R}^n$

- Many primal solvers scale with feature operations and the parameter dimension d.
- Many dual or kernel solvers scale with sample operations and sometimes with a dense Gram matrix $\mathbf{K}$ of size $n \times n$.
- The true cost depends on solver choice, sparsity, and whether dense scatter / Gram matrices are formed at all.
- Useful regime heuristic: $d \ll n$ often favors primal methods, $n \ll d$ can favor dual methods, and very large n makes dense kernel matrices expensive.





APPROXIMATE SOLUTIONS OFTEN FAVOR THE PRIMAL



- For convex problems with strong duality, the **optimal values** satisfy

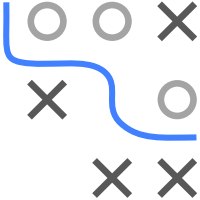
$$p^* = d^*.$$

- But this equality holds at optimality; it does **not** say that an approximately optimal dual iterate already gives a good primal model.
- In the SVM, the primal weight vector is recovered from the dual variables via

$$\boldsymbol{w}(\boldsymbol{\alpha}) = \sum_{i=1}^{n} \alpha_i y^{(i)} \boldsymbol{x}^{(i)}.$$

- A dual iterate with a good objective value can still translate to a poor primal model if the primal-dual relations are not yet accurate enough.
- By contrast, primal methods directly optimize the objective we actually care about in the original variables, so early stopping is easier to interpret.
- A dual lower bound still helps: the gap $F(\boldsymbol{w}) - L(\boldsymbol{\alpha})$ is a stopping certificate even if the main algorithm runs in the primal.

INTERPRETABILITY OF INTERMEDIATE ITERATES



Primal iterates

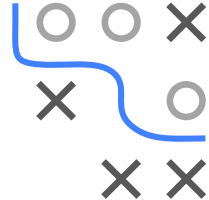
- are coefficient vectors, probabilities, portfolio weights, or separating hyperplanes
- can be inspected, constrained, or used directly during optimization
- are often meaningful even before full convergence

Dual iterates

- are multipliers, shadow prices, or point weights
- are useful diagnostically, but are one step removed from the final model
- usually need a primal recovery step before interpretation or deployment

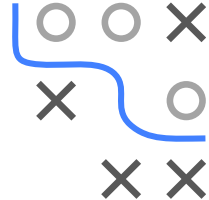
If computation stops early, a current primal iterate is usually easier to justify and communicate.

WHEN PRIMAL METHODS ARE THE DEFAULT CHOICE



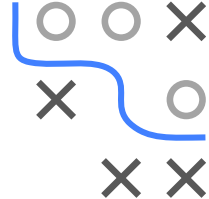
- Primal methods are usually attractive when projections, coordinate slices, or barrier terms are simple enough to compute.
- They are especially natural when d is smaller than or comparable to n, so the feature-space representation is compact.
- They are often preferable when we want interpretable parameters, coefficient bounds, sparsity patterns, or direct control over feasibility.
- They are also a good default when the primal can be written as a simple unconstrained or lightly constrained objective, as in hinge-loss or regularized risk formulations.
- In these settings, there is often little reason to pay the extra indirection of a dual reformulation.

WHEN DUAL METHODS CAN STILL BE ATTRACTIVE



- Dual methods become appealing when the number of points n is much smaller than the number of variables d.
- They can be convenient when the dual constraints are structurally simpler than the original primal constraints, often box constraints plus a few equalities.
- They are also useful when the model is naturally expressed through similarities or kernel evaluations and the matrix $\mathbf{K}$ is still manageable.
- The dual objective provides lower bounds, and the multipliers themselves can carry information about active constraints or influential observations.
- Thus, dual methods remain valuable, but only when the reformulation clearly exploits useful structure.

TAKE-AWAY



1. Primal and dual methods are two views of the same constrained optimization problem.
2. Duality is powerful, but it is not automatically better and it is not required for kernel methods.
3. Computationally, dimension gives a useful regime heuristic: $d < n$ often favors primal methods, while $n < d$ can favor dual methods.
4. For approximate solutions, primal iterates are usually easier to interpret, evaluate, and stop early.

Practical rule: use a primal method whenever it is simple enough; move to the dual only when the reformulation clearly buys something.